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Ex-11
Explore Tor Network for Anonymous Communication

AIM:

To study and understand how the Tor Network enables anonymous and secure communication over the internet by routing traffic through multiple encrypted relay nodes.

PROCEDURE:

- Download and install the Tor Browser from the official Tor Project website.
- Launch the browser and establish a connection to the Tor network.
- Browse websites using the Tor browser and observe anonymized traffic behavior.
- Observe how traffic is routed through multiple relay nodes using onion routing.
- Analyze privacy features such as IP masking, DNS leak prevention, and multi-layer encryption.

TASK-1:

Task 1 Unit 1 - Tor

Tor is a free and open-source software for enabling anonymous communication. Tor directs Internet traffic through a free, worldwide, volunteer overlay network seven thousand relays to conceal a user's location and usage from anyone conducting network surveillance or traffic analysis. Using Tor makes it more difficult t the user: this includes "visits to Web sites, online posts, instant messages, and other communication forms". Tor's intended use is to protect the personal privacy their freedom and ability to conduct confidential communication by keeping their Internet activities unmonitored.

In penetration testing, there might be a need to conduct a full-fledged black-box test. This is a form of testing in which security professionals have to deal with su other mechanisms of restriction on the customer's side. In this case, the Tor network can be used in order to constantly change IP and DNS addresses and theref overcome any restrictions.

In this unit, we are going to install the Tor service and learn basic commands.

Answer the questions below

Run `apt-get install tor` to install/update your Tor packages.

No answer needed

Run `service tor start` to start the Tor service.

No answer needed

Run `service tor status` to check Tor's availability.

No answer needed

Run `service tor stop` to stop the Tor service.

No answer needed

TASK-2:

Answer the questions below

Let's start with running `apt install proxychains` to install/update proxychains tool.

No answer needed

Now it's time to configure proxychains to work properly.

Run `nano /etc/proxychains.conf` to edit the settings. (Note: You can use any text editing tool instead of nano.)

No answer needed

We can now see, that most of the methods are under comment mark. You can read their description and decide on using one of them in the future. For this lesson `dynamic_chain` and comment others (simply put `#` to the left). Additionally, it is useful to uncomment `proxy_dns` in order to prevent DNS leak. Scroll through whenever you want to add some additional proxies at the bottom of the page (which is not required at this point).

Apply all the settings.

No answer needed

Now let's check our settings.

Start the TOR service and run `proxychains firefox`. Usually, you are required to put `proxychains` command before anything in order to force it to transfer dat

No answer needed

After the Firefox has loaded, check if your IP address has changed with any website that provides such information. Also, try running a test on `dnsleaktest.com` address changed too.

NOTE: All other web browser windows should be closed before opening Firefox through proxychains!

No answer needed

TASK-3:

Task 3 Unit 3 - Tor browser

Tor browser, as seen from its name, is a browser that transfers all its traffic through TOR and by using firefox headers makes all Tor users look the same.

On a daily basis, Tor browser is useful for anyone who wants to keep their internet activities out of the hands of advertisers, ISPs, and web sites. That includes pe
censorship restrictions in their country, police officers looking to hide their IP address or anyone else who doesn't want their browsing habits linked to them.

Install Tor browser on your system (It is not necessarily to do this on your Kali Machine).

Windows, Mac OS installation

Kali Linux guide

Answer the questions below

Finish the installation.

No answer needed

Launch the Tor Browser and set your privacy settings to Level 2 (Safer).

No answer needed

What is the search engine at the following onion address: https://duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion/

DuckDuckGo

RESULT:

The Tor network was successfully explored for anonymous communication. Traffic was encrypted and routed through multiple relay nodes (Guard, Middle, and Exit nodes), effectively hiding the user's IP address and identity. The experiment demonstrated that Tor provides a robust mechanism for preserving online privacy by preventing network surveillance and traffic analysis.